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Deepfake detection via Feature Refinement and Enhancement Network

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ABSTRACT

The rapid advancement of deepfake technology poses significant threats to the integrity and privacy of biometric systems, such as facial recognition and voice authentication. To address this issue, there is an urgent need for advanced forensic detection methods that can reliably safeguard biometric data from manipulation and unauthorized access. However, current methods mainly focus on shallow feature extraction and neglect feature refinement and enhancement, which leads to low detection accuracy and poor generalization performance. To address this problem, we propose Feature Refinement and Enhancement Network (FRENet) for deepfake detection by leveraging progressive refinement and enhanced mixed feature learning. Specifically, a Low Rank Projected Self-Attention (LPSA) module is introduced for the refinement and enhancement of features. Also, a Patch-based Focused (PatchFocus) module is proposed to highlight local texture inconsistencies in key regions. In addition, we propose a Refine Fusion (RefFus) module that integrates the refined features and associated noise information to enhance feature separability. Experimental results across five benchmark datasets demonstrate that the proposed FRENet outperforms state-of-the-art methods in terms of both accuracy and generalization. The code is available at https://github.com/weichengsong-code/FRENet.

1. Introduction

Biometric authentication, such as facial, fingerprint, and voice recognition, has become widely used in everyday smart devices. These technologies provide users with efficient and secure access to digital services. However, the growing reliance on biometric systems has also increased their exposure to sophisticated security threats [1,2]. In particular, recent advances in generative adversarial networks and diffusion models [3,4] have greatly improved the realism and diversity of facial forgeries. As a result, distinguishing between authentic and manipulated biometric data has become increasingly challenging. This raises serious concerns about the security and reliability of biometric authentication systems in the face of emerging deepfake technologies [5].

Recent studies have shown that deepfake attacks can successfully bypass state-of-the-art facial recognition systems, which pose serious threats such as unauthorized access, identity theft, and privacy invasion [6,7]. In addition, multiple reports have documented incidents of financial fraud and misinformation campaigns driven by deepfake technology, which have caused significant economic and societal harm [8]. To ensure the security and reliability of identity authentication systems, various detection techniques [9,10] have been developed. For example, Rahim et al. [11] proposed a privacy-preserving image retrieval method for mobile devices using deep features on the cloud.

Current deepfake detection methods primarily focus on either the spatial or frequency domains. In the spatial domain, Zhang et al. [12] proposed a union-saliency adversarial attack method to disrupt the functioning of deepfake generation models. This method targets the saliency regions of the generated content. It applies subtle adversarial perturbations that confuse the ability of the deepfake model to generate realistic forgeries. Xu et al. [13] introduced a reconstruction learning method that fuses gradient and content features for enhanced deepfake detection. Chen et al. [14] proposed a deepfake detection method that focuses on subtle spatial manipulation features. The model uses spatial attention to identify key forged areas and temporal consistency to recognize forgeries across frames. In the frequency domain, Luo et al. [15] proposed a multi-scale high-frequency feature extraction method to improve the generalization ability of face forgery detection across diverse datasets. Tian et al. [16] introduced a frequency-aware attentional

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feature fusion method, which captures frequency anomalies induced by image manipulations. Tan et al. [17] proposed a frequency-aware deepfake detection method to enhance generalization capabilities. However, frequency domain methods encounter challenges when processing subtle forged images, as the weakened frequency information in these images reduces their accuracy.

To further improve the accuracy rate of deepfake detection, numerous studies have adopted a combination of spatial and frequency domains. Liu et al. [18] proposed a Spatial-Phase Shallow Learning (SPSL) method to capture subtle frequency anomalies by fusing spatial and frequency domain features. This method identifies forgery traces that are often overlooked by amplitude spectrum methods [19,20]. However, SPSL directly conducts detection by fusing spatial features and frequency features, which lacks further analysis of complex features. Consequently, there is still considerable room for improvement in generalization and accuracy.

In this work, we propose a deepfake detection model termed Feature Refinement and Enhancement Network (FRENet) to improve the generalization ability by refining and enhancing the feature. FRENet integrates three key modules to support progressive feature enhancement. The Low Rank Projected Self-Attention (LPSA) module refines and enhances global features through a self-attention mechanism. The Patch-based Focused (PatchFocus) module applies localized attention to capture fine-grained texture features and subtle inconsistencies. This allows the proposed model to focus on key regions that are most susceptible to forgery. To further boost discriminative capability, the Refine Fusion (RefFus) module aggregates the refined features and associated noise information. This fusion enhances feature separability and facilitates more accurate discrimination between real and fake content. The main contributions of this paper are summarized as follows:

- We propose the FRENet model to enhance generalization by refining features and improving the model for detecting subtle forgery traces.
- We propose an LPSA module to enhance feature representation by emphasizing distinctions between real and forged content. This module guides the model to focus on subtle yet informative discrepancies. Additionally, a PatchFocus module is built to concentrate attention on key regions prone to manipulation, and it can improve the accuracy of the forgery artifact detection.
- We introduce a RefFus module that fuses the refined features and associated noise information to enhance feature separability and improve discriminative performance.

The remainder of this paper is structured as follows. Section 2 provides an overview of related work in deepfake detection. Section 3 describes the FRENet framework. Section 4 presents the experimental setup and results. Section 5 concludes the paper.

2. Related work

In recent years, deepfake detection has received increasing attention due to the rapid advancement of generative techniques. Various methods have been proposed, which can be categorized as spatial-frequency domain fusion-based methods, feature Enhancement and refinement-based methods, and local attention-based methods.

2.1. Deepfake detection based on spatial-frequency domain fusion

Early detection methods primarily focused on inconsistencies within the spatial domain and abrupt transitions in the frequency domain. For example, Rössler et al. [21] introduced a forgery detection network based on the Xception architecture. It uses depthwise separable convolutions to capture local and global patterns, which helps detect subtle deepfake artifacts. Li et al. [22] incorporated frequency-domain analysis and identified fusion boundaries in images to detect subtle artifacts.

To combine the advantages of both spatial and frequency domains, Zhai et al. [23] proposed a dual-stream frequency-spatial fusion network that combines spatial and frequency-domain information to improve deepfake detection. However, the method may incur significant computational costs due to the simultaneous processing of two distinct streams. Meanwhile, Zheng et al. [24] introduced a spatial-frequency cross-fusion model. However, the model may experience performance degradation when processing noisy or low-quality data. Yi et al. [25] proposed a deepfake detection method, termed Spatial-Frequency Feature Integration (SFIAD), to fuse spatial and frequency domain features and uses dynamic boundary optimization. While it improves detection accuracy on multiple datasets, it still faces challenges in handling highly compressed or low-quality images. Liu et al. [18] introduced the Spatial-Phase Shallow Learning (SPSL) method. The method demonstrates computational efficiency, requiring less processing power than more complex models. However, its performance remains suboptimal in complex environments. This issue is mainly attributed to limited feature extraction capabilities and a lack of comprehensive understanding of fine-grained details. Consequently, the design of a feature enhancement module is essential for refining and strengthening the generalization of the detector.

2.2. Deepfake detection based on feature enhancement and refinement

With the continuous advancement of deepfake technology, traditional detection methods face growing challenges in accurately identifying high-quality forged images. These forgeries often exhibit fewer visual artifacts and maintain strong semantic consistency. To address these limitations, recent research has increasingly focused on feature enhancement and refinement techniques. These methods aim to improve the ability of the model to capture subtle and complex differences between real and fake content. Strategies such as multi-scale feature extraction and frequency-domain representations enhance the quality and discriminability of learned features. These improvements lead to greater robustness and accuracy in deepfake detection across diverse scenarios. For example, Zhao et al. [26] proposed a multi-attentional deepfake detection network with feature enhancement to improve discriminative representation. Their method integrates multi-scale attention with enhanced feature fusion to effectively boost detection performance in complex forgery scenarios. Zhou et al. [27] proposed a frequency-aware method to improve forgery detection by fusing spatial and frequency components. Lai et al. [28] introduced a multi-feature fusion method for deepfake video detection, which extracts features from the spatial, frequency, local gravity pattern, and time domains. Qiu et al. [29] proposed the d2fusion model that enhances deepfake detection by fusing spatial and frequency domain features with attention mechanisms. The method uses feature superposition and phase updating to improve the differentiation between real and forged content. Additionally, Lin et al. [30] developed a detection method that uses convolutional neural networks to improve detection accuracy by using multiscale feature extraction. This method captures subtle inconsistencies across various levels of image detail, thereby improving the sensitivity of the model to complex forgeries.

2.3. Deepfake detection based on local attention

Attention mechanisms play a crucial role in deepfake detection. They can guide models to focus on the most informative regions of an image, thereby improving the identification of manipulated content. By assigning greater importance to salient features, attention mechanisms enhance the capacity of the model to capture subtle inconsistencies introduced during the forgery process. This targeted representation improves the accuracy and robustness of detecting both local and global artifacts.

Several studies have explored different attention-based strategies to enhance deepfake detection performance. Luong et al. [31] proposed cross-modal attention to focus on the most relevant information

across different modalities and improve the performance of deepfake detection. Han et al. [32] proposed a facial feature-guided adaptation method to promote the model to pay attention to areas most susceptible to manipulation. Waseem et al. [33] proposed a multi-attention-based approach that focuses on detecting and localizing fake faces and expression manipulations. By employing multiple attention mechanisms, the method effectively captures diverse features across different facial regions. This improves the detection accuracy of both subtle manipulations and large-scale forgeries. Lu et al. [34] proposed the Hybrid Attention Module with Attention Elimination for fine-grained image classification. This method enhances important regions and eliminates irrelevant areas to improve accuracy. Qiu et al. [35] proposed the boundary-enhanced local–global collaborative network, which combines local and global feature extraction with a boundary-focused decoder.

Overall, these approaches have achieved promising results on specific benchmarks. However, many existing methods struggle to generalize across different datasets and forgery types. Their performance often degrades in cross-domain scenarios due to overfitting or limited feature representation. This highlights the need for more robust and transferable detection models. In this context, our study introduces FRENet, a modular framework designed to enhance feature quality and improve generalization in deepfake detection.

3. Proposed method

3.1. Overall framework

The overall architecture of the FRENet method is illustrated in Fig. 1(a). The process begins with the input image being processed by PhaseNet, which reconstructs the image based on phase information extracted via Fast Fourier Transform (FFT). This phase information helps capture the underlying structure and frequency characteristics of the image. After reconstructing the image, the Inverse Fast Fourier Transform (IFFT) is applied to fuse the reconstructed image with the original input. This fusion enhances both structural and frequency-based features and improves the robustness and quality of the final image representation. The fused output is passed to the Xception network [21]. Xception employs depthwise separable convolutions and residual connections to efficiently extract high-level features. Subsequently, the features undergo refinement via the LPSA module. The primary role of LPSA is to reduce the correlation between real and fake features, which enables the model to better distinguish genuine content from manipulations.

Additionally, a PatchFocus module is incorporated to guide FRENet in focusing on local details within the image, especially those areas likely to contain manipulation artifacts. This ensures that the model focuses on regions where forgery traces are most prominent, thereby enhancing its detection capabilities. The local attention mechanism is further refined through SoftPooling operations and convolutional layers, which ensure that the focus on critical regions is accurate and efficient. The RefFus module is then used to fuse the refined features with injected noise information. This noise injection enhances the separability of features, which enables the model to more effectively distinguish between real and fake content. It functions as a regularization technique that improves the generalization of the model across diverse forgery types and varying conditions. Finally, a Multi-Layer Perceptron (MLP) classifier is employed to make the final distinction between real and fake images.

3.2. Low rank projected self-attention module

To improve generalization across diverse scenarios, we propose a Low Rank Projected Self-Attention (LPSA) module. It is illustrated in Fig. 1(c). This mechanism captures fine-grained interactions within the mixed features extracted from the input image. The input features, denoted as $F \in \mathbb{R}^{B\times C\times H\times W}$, consist of a mixture of identity and forgery features.

These mixed features are projected into a lower-dimensional embedding space using low-rank linear transformations. These transformations generate compact yet expressive query Q, key K, and value V representations:

$$Q = W_Q F, K = W_K F, V = W_V F, \tag{1}$$

where $W_Q \in \mathbb{R}^{d_q\times C}, W_K \in \mathbb{R}^{d_k\times C}, W_V \in \mathbb{R}^{d_v\times C}$, are learnable projection matrices. d_q, d_k, and d_v represent the dimensions of the query, key, and value embeddings, respectively.

Subsequently, the self-attention is computed to capture the correlations across different semantic aspects of the features. Specifically, the attention scores quantify how significantly each region in artifact features influences the corresponding region in identity features. The self-attention operation is formulated as:

$$F_S = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V, \tag{2}$$

where the scaling factor $\sqrt{d_k}$ stabilizes gradients during training. This operation aggregates relevant artifact information to enrich the identity representations. It strengthens the discriminative capability of the features by capturing subtle manipulation traces. These include texture inconsistencies, boundary artifacts, and frequency abnormalities. As a result, the model gains an improved ability to distinguish real content from forgeries. The enriched features also enhance the robustness of the model under varied forgery conditions and unseen data.

3.3. Patch-based focused module

To effectively highlight informative local features, we introduce a local attention mechanism specifically tailored for deepfake detection tasks. The architecture of this mechanism is illustrated in Fig. 2. Distinct from traditional pooling methods, the local attention module adopts a Patch strategy that adaptively assigns local weights. This mechanism enables the model to concentrate on spatially important regions with greater precision.

The input features are projected through a 1×1 convolution into a compact representation. This projection reduces both the dimensionality and the computational complexity. Subsequently, the Patch operation aggregates local information using exponential weighting. This approach ensures smooth feature transitions and facilitates the capture of discriminative spatial patterns.

The local attention module is represented as:

$$W = \sigma(Conv_{3\times3}(P(Conv_{1\times1}(F_S))), \tag{3}$$

where F_S denotes the input feature maps, $\sigma(\cdot)$ indicates the sigmoid activation, and $P(\cdot)$ is defined explicitly as:

$$P(F_S) = \frac{\text{SoftPool}(F_S \cdot e^{F_S})}{\text{SoftPool}(e^{F_S})}, \tag{4}$$

where SoftPool$(\cdot)$ refers to a soft pooling operation based on average pooling in a local window. The term e^{F_S} acts as an exponential attention weight. This design increases the influence of features with larger values. As a result, $P(F_S)$ highlights distinctive local characteristics and promotes the selection of informative features. The network can thus focus more on important regions in the feature maps.

In addition, we incorporate an auxiliary gating mechanism to further refine the attention. The gating mechanism adaptively recalibrates the importance of the features by selectively emphasizing critical spatial locations while attenuating redundant or noisy information. Specifically, the gating function utilizes the first feature channel to generate a refined importance map. This approach leverages the inherent representative strength of the channel as a simplified yet powerful gating reference. The gating operation is formulated as follows:

$$g = SN(Conv_{1\times1}(F_S[:, 1, :, :])), \tag{5}$$

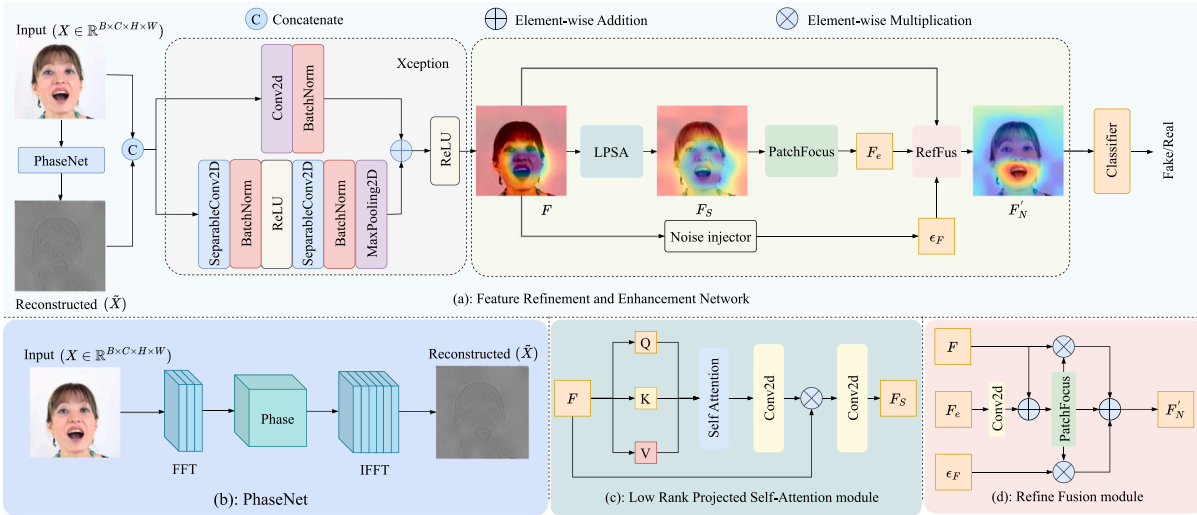

Fig. 1. Overview of the FRENet model for deepfake detection.

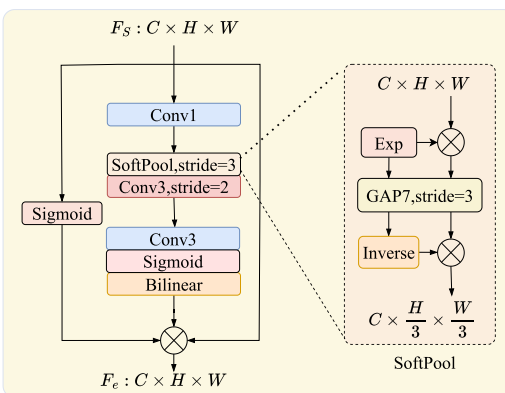

Fig. 2. Framework of the PatchFocus module.

where $Conv_{1\times1}(\cdot)$ performs a convolution to refine the feature map, and $SN(\cdot)$ provides a smooth nonlinear mapping to constrain the gating values between 0 and 1. This adaptive gating not only suppresses irrelevant details but also dynamically emphasizes pertinent information critical for distinguishing genuine and forged patterns. Consequently, the refined attention feature maps are obtained by integrating the attention weight and gating factor g with the original input F_s. The output formula of the PatchFocus module is expressed as:

$$F_e = F_S \cdot W \cdot g, \tag{6}$$

The proposed local attention module incorporates a combined patch-based mechanism and an enhanced gating strategy. This design enables the model to emphasize subtle yet critical local differences that are highly relevant to deepfake detection. By focusing on these fine-grained local variations, the module captures manipulation cues that may be overlooked by global feature extractors. As a result, the discriminative power of the learned features is significantly improved. Additionally, the enhanced attention to local inconsistencies contributes to greater model robustness across various forgery types and manipulation scenarios.

3.4. Refine fusion module

The RefFus module employs residual connections to integrate refined features, followed by the injection of noise information to improve generalization across diverse forgery scenarios. The overall framework is illustrated in Fig. 1(d). It enhances the ability of the model to capture the complex relationships between identity and forged artifact features. Residual connections help alleviate the problem of vanishing gradients and facilitate better information flow during training.

In the RefFus module, convolutional residual connections are applied to the refined features after the self-attention operation. The residual convolution operation is formulated as follows:

$$F_e' = F + Conv_{1\times1}(F_e), \tag{7}$$

where $Conv_{1\times1}(\cdot)$ represents a convolutional operation. The 1×1 convolution operation on an input feature map $F_e \in \mathbb{R}^{C\times H\times W}$ is formulated as:

$$Conv_{1\times1}(F_e) = W_c \cdot F_e + b_c, \tag{8}$$

where $W_c \in \mathbb{R}^{C'\times C\times1\times1}$ denotes the convolution kernel, $b_c \in \mathbb{R}^{C'}$ is the bias term. Specifically, the 1×1 convolution serves two main purposes. First, it adaptively recalibrates channel-wise feature representations, thereby enhancing the ability of the model to capture salient information. Second, it provides dimensionality preservation and feature transformation without changing spatial dimensions, thereby facilitating effective fusion of residual information. The final residual-enhanced features F_e' is formulated as:

$$F_e'(c,h,w) = F_e + \sum_{c'=1}^{C} W_c(c,c',1,1) \cdot F(c',h,w) + b_c(c), \tag{9}$$

where c indicates the channel index, and (h,w) are spatial coordinates within the feature map.

To enhance the robustness and generalization of the proposed method, Gaussian noise is injected into the refined features to obtain mixed noise characteristics F_N. This enables the model to prevent overfitting and increases its resilience to subtle feature variations typically encountered in facial authentication scenarios. Specifically, the noise-equipped identity features are computed as follows:

$$F_N = (1 - F_e') \odot F + F_e' \odot \epsilon_F, \tag{10}$$

where $\epsilon_F \sim N(\mu_{X_F}, \sigma^2_{X_F})$ represents Gaussian noise. Here, $\odot$ denotes the element-wise product. The introduced noise effectively perturbs the feature maps, which prevents the model from focusing excessively on dataset-specific characteristics or irrelevant details and promotes robustness while improving generalization across unseen datasets.

The enhanced features undergo an additional convolutional step to consolidate their representations into more discriminative embeddings

suitable for the final classification stage. The corresponding expression is formulated as follows:

$$F'_N = Conv_{1\times1}(F_N). \tag{11}$$

The proposed method includes a structured feature fusion and noise injection module. This design strengthens its ability to distinguish between real and fake features. The module integrates refined feature representations with controlled noise signals. This process enhances the separability between authentic and manipulated content. The injected noise acts as a regularization factor and helps reduce overfitting to specific patterns. It also improves the generalization of the model to various types of forgery.

4. Experimental results and analysis

4.1. Implementation details

In this work, we use the Adam [36] optimizer and configure the optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1\times10^{-8}$, and a weight decay of 0.0005. The learning rate is set to 2×10^{-4} and the batch size is 32. We resize all input images to 256 × 256 pixels and normalize the images using a mean and standard deviation of $[0.5, 0.5, 0.5]$. The backbone is the official Xception model. We initialize the model weights with parameters pre-trained on the ImageNet dataset and adopt cross-entropy loss as the objective function. Several data augmentation strategies are adopted, for example, random horizontal flip with a probability of 0.5, random rotation in the angle range of $[-10°, 10°]$, and random Gaussian blur with a probability of 0.5. The JPEG compression quality is uniformly sampled between 40 and 100. All experiments are carried out using the PyTorch library [37]. We conduct the training and evaluation on two NVIDIA 3090 Ti GPUs. The training runs for 10 epochs. We set the random seed to 1024 to ensure reproducibility.

4.2. Datasets

The experiments were trained in the FaceForensics ++ (FF ++) [21] dataset and evaluated on five benchmark datasets, which include FF++[21], Celeb-DF-v1 [38], Celeb-DF-v2 [38], DFDCP [39], and FaceShifter [40].

The FF++ dataset is a comprehensive resource for training and evaluating deep learning models in facial manipulation detection. It contains 1000 original videos which includes 4000 forged videos generated using four representative facial forgery techniques, namely DeepFakes (DF), Face2Face (F2F), FaceSwap (FS), and NeuralTextures (NT). In total, the dataset contains over 1.8 million manipulated images, which are extracted video frames.

The Celeb-DF-v1 dataset contains 795 real videos featuring 59 celebrity subjects, along with 5639 high-quality deepfake videos. This dataset incorporates early deepfake generation methods, such as neural network-based facial synthesis and blending.

The Celeb-DF-v2 provides high-resolution videos with an equal distribution of real and fake content, generated using advanced deepfake techniques. This makes the dataset a valuable benchmark for testing and improving deepfake detection methods.

The DFDCP dataset comprises high-quality real and manipulated videos generated using diverse facial forgery techniques. It emphasizes realism by including variations in lighting conditions, camera angles, and video resolutions. These characteristics render the dataset suitable for training and benchmarking robust deepfake detection models under practical conditions.

The FaceShifter dataset provides highly realistic face-swapping examples with minimal visual artifacts. It ensures consistency in lighting conditions, background alignment, and facial identity preservation.

4.3. Evaluation metrics

The proposed model is evaluated using four standard binary classification metrics, namely Area Under the Curve (AUC), Accuracy (ACC), Average Precision (AP), and Equal Error Rate (EER). These metrics provide a comprehensive assessment of the detection performance of the model from different perspectives. The AUC measures the ability of the model to distinguish between real and fake samples. It reflects the probability that the model correctly ranks a randomly chosen positive sample higher than a negative one. A higher AUC value indicates better separability between classes. The ACC quantifies the proportion of correctly classified samples over the total number of samples. It provides an overall measure of the effectiveness of the model in detecting deepfakes. The AP evaluates the trade-off between precision and recall across various decision thresholds. It is particularly useful for assessing the performance of the model in imbalanced classification scenarios. The EER represents the point at which the false acceptance rate equals the false rejection rate. A lower EER indicates better balance and reliability in classification, especially in biometric and security-related tasks.

4.4. Experiments and discussion

4.4.1. Cross-testing

To evaluate the generalization and robustness of the proposed FRENet approach, we conducted cross-dataset testing. The model was trained on the FF++ dataset and tested on four additional datasets, namely Celeb-DF-v1, Celeb-DF-v2, DFDCP, and FaceShifter. These datasets provide a diverse range of deepfake content, which ensures that the evaluation covers a variety of forgery types, generation methods, and image qualities. The detailed results and comparisons are presented in Table 1. It proves that the FRENet outperforms other competitors in the four datasets. Specifically, compared to the baseline model, SPSL, the FRENet improves the AUC by 8.56%, 5.1%, 3.62%, 10.5% in the Celeb-DF-v1, Celeb-DF-v2, DFDCP, and FaceShifter, respectively. Meanwhile, it achieves an average increase of 5. 13% compared to SPSL. Additionally, some competing methods show significant decreases in certain datasets. In contrast, FRENet maintains more stable AUC scores on all benchmarks. This stability indicates a strong robustness against domain shift and various forgery techniques.

To further validate the superiority of the proposed method, the FRENet model was trained on the FF++ dataset and evaluated on four widely recognized benchmark datasets, namely Celeb-DF-v1, Celeb-DF-v2, DFDCP, and FaceShifter. Three key evaluation metrics, ACC, AP, and EER, were employed. These metrics provide a comprehensive assessment of the performance of the model in terms of both classification accuracy and its ability to distinguish between authentic and manipulated content. The results are presented in Table 2. It shows that FRENet consistently outperforms the baseline model across all evaluation metrics on the four datasets. In particular, the improvements in ACC, AP, and EER further emphasize the effectiveness of FRENet in detecting facial forgeries, regardless of the dataset or the specific manipulation techniques employed. This reinforces the robustness and generalization of the model, which confirms that FRENet is a highly effective solution for deepfake detection across a wide range of datasets and conditions.

4.4.2. Intra-testing

The intra-dataset evaluation demonstrates the overall effectiveness and stability of FRENet across multiple types of deepfake generation techniques. Table 3 provides a detailed comparison of FRENet with several state-of-the-art deepfake detection methods under intra-dataset settings. All models are trained and tested on the FF++ dataset and its subsets, including DF, F2F, FS, and NT. FRENet achieves the highest AUC scores in the FS and F2F subsets, exceeding the next-best method by 0.09% and 0.16%, respectively. These results highlight

Table 1
Comparative results on the cross-testing experiment. The training dataset is the FF++ dataset, and the metric is AUC. The best results are highlighted in **bold**.

Methods	Venue/Year	Celeb-DF-v1	Celeb-DF-v2	DFDCP	FaceShifter	Avg.
MesoIncep [41]	IEEE WIFS 2018	0.7366	0.6966	0.7561	0.6438	0.7083
EfficientB4 [42]	PMLR 2019	0.7909	0.7487	0.7283	0.6162	0.7210
Capsule [43]	IEEE 2019	0.7909	0.7472	0.6568	0.6465	0.7104
FWA [44]	CVPRW 2019	0.7897	0.6680	0.6375	0.5551	0.6626
CNN-Aug [45]	CVPR 2020	0.7420	0.7027	0.6170	0.5985	0.6651
FFD [46]	CVPR 2020	0.7840	0.7435	0.7426	0.6056	0.7189
F3Net [47]	ECCV 2020	0.7769	0.7352	0.7354	0.5914	0.7097
SPSL [18]	CVPR 2021	0.8150	0.7650	0.7408	0.6437	0.7411
SRM [15]	CVPR 2021	0.7986	0.7552	0.7408	0.6014	0.7225
CORE [48]	CVPR 2022	0.7798	0.7428	0.7341	0.6032	0.7150
Recce [49]	CVPR 2022	0.7677	0.7319	0.7419	0.6095	0.7128
VGG19 [50]	IEEE BigData 2024	0.7931	0.7667	0.7076	–	–
ADA-FInfer [51]	IEEE TDSC 2024	–	0.7610	–	–	–
SFIAD [25]	Artif. Intell. Rev. 2025	0.8840	0.7900	0.7588	–	–
SFE-Net [52]	IEEE ICASSP 2025	0.8660	0.7980	0.7600	–	–
CAFMF [53]	IEEE ICCRD 2025	0.8090	0.7758	–	0.6575	–
FRENet (Ours)	–	**0.8848**	**0.8040**	**0.7696**	**0.7114**	**0.7924**

Table 2
Comparative results between the FRENet and baseline (SPSL). The training dataset is the FF++ dataset, and the metrics are ACC, AP, and EER. The best results are highlighted in **bold**.

	Celeb-DF-v1			Celeb-DF-v2			DFDCP			FaceShifter		
	ACC↑	AP↑	EER↓	ACC↑	AP↑	EER↓	ACC↑	AP↑	EER↓	ACC↑	AP↑	EER↓
Baseline	0.7091	0.8910	0.2543	0.6686	0.8555	0.2998	0.6876	0.8214	0.3341	0.6392	0.6822	0.3549
FRENet (Ours)	**0.7630**	**0.9214**	**0.1862**	**0.7465**	**0.8776**	**0.2802**	**0.7106**	**0.8492**	**0.3053**	**0.6412**	**0.6868**	**0.3471**

Table 3
Comparative results on the intra-testing experiment. The training dataset is the FF++ dataset, and the metric is AUC. The best and second-best results are highlighted in **bold** and underlined, respectively.

Methods	Venue/Year	FF++	DF	F2F	FS	NT	Avg.
MesoIncep [41]	IEEE WIFS 2018	0.7583	0.8542	0.8087	0.7421	0.6517	0.7630
EfficientB4 [42]	PMLR 2019	0.9567	0.9757	0.9758	0.9797	0.9308	0.9637
Capsule [43]	IEEE 2019	0.8421	0.8669	0.8634	0.8734	0.7804	0.8452
FWA [44]	CVPRW 2019	0.8765	0.9210	0.9000	0.8843	0.8120	0.8788
CNN-Aug [45]	CVPR 2020	0.8493	0.9048	0.8788	0.9026	0.7313	0.8534
F3Net [47]	ECCV 2020	0.9635	0.9793	0.9796	0.9844	0.9354	0.9684
FFD [46]	CVPR 2020	0.9624	0.9803	0.9784	0.9853	0.9306	0.9674
SPSL [18]	CVPR 2021	0.9610	0.9781	0.9754	0.9829	0.9299	0.9655
SRM [15]	CVPR 2021	0.9576	0.9733	0.9696	0.9740	0.9295	0.9609
CORE [48]	CVPR 2022	0.9638	0.9787	0.9803	0.9823	0.9339	0.9678
Recce [49]	CVPR 2022	0.9621	0.9797	0.9779	0.9785	0.9357	0.9668
VGG19 [50]	IEEE BigData 2024	–	0.9700	0.9641	0.9769	0.9135	–
ADA-FInfer [51]	IEEE TDSC 2024	–	0.8310	0.7840	0.7440	0.7230	–
SDIF [54]	ICASSP 2024	**0.9993**	0.8354	0.8369	0.7514	–	–
DDL [55]	IEEE TIFS 2025	0.7090	0.8310	0.7750	0.7810	0.7930	0.7778
CAFMF [53]	IEEE ICCRD 2025	–	**0.9881**	0.9817	0.9894	**0.9439**	–
FRENet (Ours)	–	0.9688	0.9856	**0.9833**	**0.9903**	0.9376	**0.9731**

the strong ability of FRENet to detect face-swapping and face-to-face manipulations. In the DF and NT subsets, the AUC scores of FRENet are slightly lower than the best-performing methods. For the complete FF++ dataset, FRENet achieves an AUC that is also slightly lower than the top-performing method.

Compared to the SPSL baseline, FRENet improves the average AUC by 0.76%. While some competing methods, such as SDIF and CAFMF, achieve high results in individual subsets, they do not consistently maintain strong performance across all datasets. In contrast, FRENet consistently provides high and stable performance, regardless of the type of manipulation. Certain methods, such as DDL and Mesonecp, experience notable performance drops in challenging subsets, which limits their adaptability. FRENet avoids such declines and provides reliable detection under all evaluated conditions. These findings indicate that FRENet not only achieves state-of-the-art accuracy but also delivers practical and stable performance for deepfake detection across a wide range of manipulation techniques.

4.4.3. Visualized analysis

To provide further visual evidence of the effectiveness of the FRENet, the subjective visualization results are shown in Fig. 3. As illustrated in the figure, the proposed method demonstrates a strong capability to distinguish between real and fake images across multiple datasets. The visualizations highlight the regions where FRENet detects discrepancies, which showcase its ability to accurately identify manipulated content. This confirms the robustness and effectiveness of the method in handling various deep-fake generation techniques.

In addition, the attention map visualizations are presented in Fig. 4. These visualizations provide insight into the regions that the model prioritizes during the detection process. Compared to the baseline SPSL model, FRENet focuses more on facial regions that are commonly vulnerable to forgeries. Specifically, the model highlights areas such as the eyes, mouth, and facial contours. This targeted attention improves the detection of subtle manipulations across different datasets.

Overall, the attention maps demonstrate that FRENet can detect more potential forgery traces than the baseline model. By concentrating

Fig. 3. Visualization of qualitative results. Ground truth (GT) and prediction (Pred) are highlighted in red and blue, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

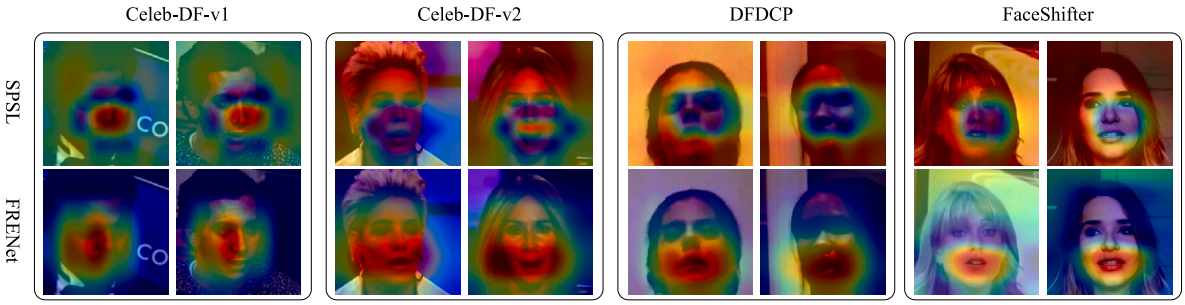

Fig. 4. Visualizations of the attention map between the proposed FRENet and the baseline (SPSL).

Table 4
The ablation studies of the core modules. The best performance for each metric is indicated in **bold**.

Methods	AUC↑	ACC↑	AP↑	EER↓
Baseline	0.8192	0.7097	0.8910	0.2543
Baseline + LPSA	0.8573	0.6875	0.9029	0.2236
Baseline + RefFus	0.8505	0.7413	0.9029	0.2186
Baseline + LPSA + RefFus	0.8559	0.7302	0.9071	0.2161
Baseline + RefFus + PatchFocus	0.8417	0.7248	0.8974	0.2360
Baseline + LPSA + RefFus + PatchFocus	**0.8848**	**0.7630**	**0.9214**	**0.1862**

on critical facial areas, FRENet identifies forgery signs that might be overlooked by other methods. This improvement in detection capability further emphasizes the superiority of FRENet in the domain of deepfake detection.

4.4.4. Ablation study

To evaluate the effectiveness of each key component of the proposed model, we conducted ablation experiments on the Celeb-DF-v1 dataset. In these experiments, the baseline model for comparison is SPSL [18]. The main components analyzed include LPSA, PatchFocus, and RefFus. LPSA serves as the feature refinement module, which reduces the correlation between real and fake features and thus enhances discriminability. PatchFocus acts as a local attention module that guides the model to focus on regions prone to forgery. RefFus is the feature enhancement module, designed to fuse refined features with auxiliary information to improve feature separability and robustness. The results of the ablation experiment are presented in Table 4.

It shows that the introduction of the single LPSA module improves the performance of AUC, AP and EER, but the ACC decreases slightly. This outcome indicates that the LPSA module improves the ability to distinguish between real and fake images, while the classification accuracy shows minor fluctuation. When the RefFus module is independently equipped, the four indicators are all improved to certain extent. It demonstrates that RefFus enhances both detection and classification performance. When LPSA and RefFus are both equipped, the improvements are more evident. The indicators of AUC, ACC, and AP are enhanced by 3.67%, 2.05%, and 1.61%, respectively. Meanwhile, the EER is decreased by 3.82% compared to baseline. The combination of PatchFocus and RefFus yields moderate improvements. The AUC, ACC, and AP are increased 2.25%, 1.51%, and 0.64%, respectively, and EER is decreased by 1.83%. These improvements are less significant than those obtained with the combination of LPSA and RefFus. Overall, the integration of LPSA, RefFus, and PatchFocus achieves the best performance. The AUC, ACC, and AP are increased by 6.56%, 5.33%, and 3.04%, respectively, and EER is decreased by 6.81% relative to baseline. These ablation results demonstrate that each proposed module contributes significantly to the overall performance of the model. The modular design leads to consistent and substantial improvements in deepfake detection, enhancing both robustness and accuracy under varied forgery conditions.

4.4.5. Hyperparameter study

This section investigates the influence of the learning rate and the weight decay on the performance of the model. Table 5 presents the results of varying these hyperparameters. The learning rate values include 0.0001, 0.0002, and 0.0003. The weight decay values include 0.0004, 0.0005, and 0.0006. The experimental results show that hyperparameter selection directly affects the model AUC. FRENet achieves the best performance with a learning rate of 0.0002 and a weight decay of 0.0005, where the AUC reaches 88.48%. In comparison, both lower and higher values of these hyperparameters result in reduced performance. Based on the experimental findings, subsequent experiments adopt the best-performing learning rate and weight decay values.

5. Conclusion

This work proposes a deepfake detection model termed FRENet. The model uses a modular architecture to improve the generalization

Table 5
Comparative results for different learning rates and weight decay values using FRENet. The best results are highlighted in **bold**.

	Learning Rate			Weight Decay		
	0.0001	0.0002	0.0003	0.0004	0.0005	0.0006
AUC	0.8260	**0.8848**	0.8526	0.8347	**0.8848**	0.8267

of the network. FRENet refines and enhances features in multiple stages. The model captures subtle and complex traces of forgery. This process strengthens the ability to distinguish real content from manipulated content. Experimental results show that FRENet surpasses several state-of-the-art methods on multiple benchmark datasets. The model demonstrates improved generalization across different forgery types and datasets. These results confirm its potential for real-world applications. The approach also contributes to more reliable and scalable solutions for digital media authentication.

Despite these strengths, some limitations exist. FRENet mainly addresses image-based deepfake detection. This focus may reduce its performance with multi-modal forgeries. The model also shows considerable computational complexity. This feature limits deployment in real-time systems and on resource-limited devices. FRENet performs well on standard benchmarks, but further tests are needed for unseen or advanced forgery techniques. Future work will optimize FRENet for efficiency in real-time settings. The researchers plan to develop lightweight versions for edge deployment. The extension of the method to video and multi-modal deepfake detection represents another research direction. The use of self-supervised or unsupervised learning methods may further improve adaptability. These strategies can help the model handle new forgery techniques without large amounts of labeled data.

CRediT authorship contribution statement

Weicheng Song: Methodology. **Siyou Guo:** Writing – original draft. **Mingliang Gao:** Supervision. **Qilei Li:** Software, Resources. **Xianxun Zhu:** Supervision. **Imad Rida:** Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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